MATH317 · NUMERICAL METHODS

Midterm 471 Solutions

20/10/2025 · 60 minutes · 30 marks

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01 Solution Q1: errors

Exact value:	$r = x - y = 0.00016$	[1 mark]
Round to four digits:	$\tilde{x} = 0.5462$ $\tilde{y} = 0.5460$	[1 mark]
Approximation:	$\tilde{r} = 0.5462 - 0.5460 = 0.0002$	[1 mark]
Absolute error =	$\lvert 0.00016 - 0.0002 \rvert = 0.00004$	[2 marks]

Relative error $= \dfrac{0.00004}{0.00016} = 0.25$ [2 marks]

The relative error is 25% because two nearly equal numbers were subtracted.

02 Solution Q2: Horner and Taylor

Coefficients: `3, −4, 0, −12`. The zero stands for the missing `x` term.

a)	3	−4	0	−12
−2 |		−6	20	−40
	3	−10	20	−52

$P(-2) = -52$

b)	3	−4	0	−12	
3 |		9	15	45	
	3	5	15 |	33	c0 = 33
		9	42		
	3	14 |	57		c1 = 57
		9			
	3 |	23			c2 = 23
	3				c3 = 3

$$P(x) = 3(x-3)^3 + 23(x-3)^2 + 57(x-3) + 33$$

03 Solution Q3: Floating-point

1. $(45)_{10} = (55)_8 = (101101)_2$

2. $(.21875)_{10} = (.16)_8 = (.001110)_2$

3. $(45.21875)_{10} = (101101.001110)_2 = (1.01101001110)_2 \times 2^5$

   stored mantissa: .01101001110

4. c − 127 = 5 → c = 132 = $(204)_8$ = $(1000\ 0100)_2$

5. $[1\ 10000100\ 01101001110...0000]_2$

   $[1100\ 0010\ 0011\ 0100\ 1110\ 0000\ ...\ 0000]_2$

   $= [C234E000]_{16}$

A note on the exam paper

The question text says "single-precision ... in a 64-bit word-length", which is a typo. The key solves it as single precision (32-bit).

04 Solution Q4: Bisection

a) $u = f(2) = 8 - 18 + 1 = -9$ $v = f(4) = 64 - 36 + 1 = 29$

$u \cdot v < 0 \rightarrow$ f has a root α in $[2, 4]$

$f'(x) = 3x^2 - 9 > 0$ on $[2, 4] \rightarrow$ f is increasing $\rightarrow$ exactly one root

b) $n = 0$: $x_0 = \frac{2 + 4}{2} = 3$ $w = f(3) = 1$

$u \cdot w < 0 \rightarrow b = x_0 \rightarrow [2, 3]$

$n = 1$: $x_1 = \frac{2 + 3}{2} = 2.5$ $w = f(2.5) = -5.875$

$u \cdot w > 0 \rightarrow a = x_1 \rightarrow [2.5, 3]$

$n = 2$: $x_2 = \frac{2.5 + 3}{2} = 2.75$ $w = f(2.75) = -2.9531$

$u \cdot w > 0 \rightarrow a = x_2 \rightarrow [2.75, 3]$

n	x_n	$f(x_n)$	Error
0	3	1	1
1	2.5	−5.875	0.5
2	2.75	−2.9531	0.25

Notes on the key

The key writes `f(4) = 38`. The correct value is `64 − 36 + 1 = 29`. The conclusion is unchanged because the sign is positive either way.

The key ends with the note: skipping any step will result in losing a grade.